

FIT9132 Applied Week 4

A4-1 The Relational Model

A41.1 Relational Model Theory

- 1) **Relation** : A set of attributes consists of a head and a body,
The heading is the schema, and the body is the sets of tuples.
- 2) **Attribute** : A characteristic of an objective that we want to record
- 3) **Domain** : A set of atomic values from which attribute's value are drawn. Domain consists of a name, data type and data format. For example: cust_age: NUMBER(3)
- 4) **Tuple** : A set of related values that describe an instance of the relation (It is a row/ observation)
- 5) **Degree and Cardinality of a relation** :
 - a) **Degree** : Number of attributes
 - b) **Cardinality** : Number of Tuples
- 6) **Primary Key, Super Key and Foreign Key**
 - a) **Super Key** : A set of attributes that uniquely define a tuple.
 - b) **Primary Key** : Unique identifier (An attribute or a MINIMAL set of attributes that uniquely identifies each tuple).
 - c) **Foreign Key** : An attributes in a relation that exists in the same or another relation as Primary Key.

A4-1.2 Choosing the Primary Key

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- 2) Identify the primary key and foreign key for these three relations:
ORDER (order_id, order_date, cust_id)
Primary Key : order_id
Foreign Key : None

ORDERLINE (order_id, prod_no, ol_qtyordered, ol_lineprice)
Primary Key : order_id
Foreign Key : order_id, prod_no

PRODUCT (prod_no, prod_desc, prod_unitprice)
Primary Key : prod_no
Foreign Key : None

3) APPOINTMENT (dentist_id, dentist_name, patient_id, patient_name, appointment_datetime, surgery_roomno)

Identify the superkey(s), candidate key(s) and the primary key for the relation if the following business rules are applicable:

- a) A dentist can only see a single patient at a particular date and time
- b) A dentist treats a patient in a particular surgery room (only one patient is treated in the room at a time), and
- c) A patient can see the same dentist multiple times.

Candidate Key :

dentist_id, app_datetime

patient_id, app_datetime

sur_roomno, app_datetime

Primary Key :

Dentist_id

Super Key :

dentist_id, appointment_datetime (because of rule 1)

surgery_roomno, appointment_datetime (because of rule 2)